LICONG LIN

PERSONAL INFORMATION

Email: liconglin@berkeley.edu
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Address: 437 Evans Hall, UC Berkeley, Berkeley, CA 94720

EDUCATION

University of California, Berkeley 08/2021 - Present
Ph.D. candidate in Statistics; GPA: 4.0/4.0
Advisors: Song Mei and Peter Bartlett

Peking University 09/2017 - 07/2021
B.S. in Statistics; graduated with honors
GPA: 3.82/4.00; rank: 2/45

RESEARCH INTERESTS

- Theoretical foundations of AI, AI alignment, reinforcement learning, high-dimensional statistics.
- My work focuses on the **statistical foundations and algorithms for AI**. I use and extend tools from statistics and optimization to study the theoretical foundations of models, algorithms and phenomena in AI (e.g., Transformers, reinforcement learning, contrastive learning, scaling laws), and to build mathematically motivated algorithms for AI alignment (e.g., LLM unlearning).

WORK EXPERIENCE

- Applied scientist intern, **Amazon Web Services**; Santa Clara, CA 05/2024 - 08/2024
  - Research on accelerating LLM inference via speculative decoding.
  - Proposed approximate speculative decoding algorithms that optimally trade off sampling accuracy and decoding speed.
- Student researcher, **Google Research**; New York, NY 01/2025 - 05/2025
  - Research on the theoretical foundations of generative retrieval.
  - Analyzed the expressive power of score-based retrieval and generative retrieval methods.
- Quantitative research intern, **The Voleon Group**; Berkeley, CA 06/2025 - 08/2025

SELECTED PAPERS

- Ruiqi Zhang*, **Licong Lin***, Yu Bai, Song Mei. "Negative preference optimization: From catastrophic collapse to effective unlearning." *Conference on Language Modeling (COLM)* (2024). arXiv.
- **Licong Lin**, Yu Bai, and Song Mei. "Transformers as decision makers: Provable in-context reinforcement learning via supervised pretraining." *International Conference on Learning Representations (ICLR)* (2024). arXiv.
- Kazusato Oko*, **Licong Lin***, Yuhang Cai*, and Song Mei*. "A statistical theory of contrastive pre-training and multimodal generative AI." *arXiv preprint:2501.04641* (2025), *major revision at Journal of the American Statistical Association (JASA)*. arXiv.

HONORS AND AWARDS

- Outstanding Graduate Student Instructor Award, UC Berkeley — 2023
- Huaixin Bachelor, Peking University — 2021
- Honor graduate of Applied Mathematics and Statistics Program, Peking University — 2021
- Academic Excellence Award, Peking University — 2018, 19, 20
- Peking University Scholarship, Peking University — 2019, 20
- Gold Medal in Probability & Statistics, S-T. Yau College Student Mathematics Contest, placed **1st nationally**, China — 2020
- 1st Prize, Beijing College Student Mathematics Competition, China — 2018
- 2nd Prize, China National Mathematical Olympiad, China — 2016

SKILLS

Programming: Python, R, Matlab, Git, LaTeX.

TEACHING EXPERIENCE

Graduate Student Instructor (TA), **UC Berkeley**; Berkeley, CA — 01/2022 - 12/2023

- STAT 153: Introduction to Time Series (Instructor: Ruoqi Yu, Spring 2022).
- STAT 210B: Theoretical Statistics II (Instructor: Song Mei, Spring 2023).
- STAT 135: Concepts of Statistics (Instructor: Adam Lucas, Fall 2023).

Received UC Berkeley Outstanding Graduate Student Instructor Award in 2023.

SERVICES

- **Journal Reviewer**: Annals of Statistics (AoS); Journal of the American Statistical Association (JASA); SIAM Journal on Mathematics of Data Science (SIMODS); Journal of Machine Learning Research (JMLR); Transactions on Machine Learning Research (TMLR).
- **Conference Reviewer**: International Conference on Machine Learning (ICML); Conference on Neural Information Processing Systems (NeurIPS); Conference on Learning Theory (COLT); International Conference on Learning Representations (ICLR); International Conference on Artificial Intelligence and Statistics (AISTATS); IEEE Information Theory Workshop (ITW).

  Received Top Reviewer Award at NeurIPS 2025.
- **Mentorship**: Student mentor for the Statistics Undergraduate Mentorship (SUM) program and Statistics Graduate Student Association (SGSA) mentorship program, UC Berkeley, 2022–2025.

PREPRINTS

($\star$ for co-first author or alphabetical order)

6. Xuelin Yang, **Licong Lin**, Susan Athey, Michael I. Jordan, and Guido W. Imbens. "Cross-Validated Causal Inference: a Modern Method to Combine Experimental and Observational Data." *arXiv preprint:2511.00727* (2025). arXiv.
5. Kazusato Oko*, **Licong Lin***, Yuhang Cai*, and Song Mei*. "A statistical theory of contrastive pre-training and multimodal generative AI." *arXiv preprint:2501.04641* (2025), *major revision at Journal of the American Statistical Association (JASA)*. arXiv.

4. Ruiqi Zhang, Jingfeng Wu, **Licong Lin**, and Peter L. Bartlett. "Minimax optimal convergence of gradient descent in logistic regression via large and adaptive stepsizes." *arXiv preprint:2504.04105* (2025), an earlier version was accepted at ICML 2025, *minor revision at Journal of Machine Learning Research (JMLR)*. arXiv.

3. **Licong Lin***, Fangzhou Su*, Wenlong Mou, Peng Ding, and Martin J. Wainwright. "When is it worthwhile to jackknife? Breaking the quadratic barrier for Z-estimators." *arXiv preprint:2411.02909* (2024). arXiv.

2. Michael Celentano*, Zhou Fan*, **Licong Lin***, and Song Mei*. "Mean-field variational inference with the TAP free energy: Geometric and statistical properties in linear models." *arXiv preprint:2311.08442* (2023). arXiv.

1. Taejoo Ahn*, **Licong Lin***, and Song Mei*. "Near-optimal multiple testing in Bayesian linear models with finite-sample FDR control." *arXiv preprint:2211.02778* (2022). arXiv.

PUBLICATIONS

11. **Licong Lin**, Jingfeng Wu, and Peter L. Bartlett. "Improved scaling laws in linear regression via data reuse." *Advances in Neural Information Processing Systems (NeurIPS)* (2025). arXiv.

10. **Licong Lin** and Song Mei. "A statistical theory of contrastive learning via approximate sufficient statistics." *Advances in Neural Information Processing Systems (NeurIPS)* (2025). arXiv.

9. Chongyu Fan*, Jiancheng Liu*, **Licong Lin***, Jinghan Jia, Ruiqi Zhang, Song Mei, and Sijia Liu. "Simplicity prevails: Rethinking negative preference optimization for LLM unlearning." *Advances in Neural Information Processing Systems (NeurIPS)* (2025). arXiv.

8. **Licong Lin***, Koulik Khamaru*, and Martin J. Wainwright. "Semiparametric inference based on adaptively collected data." *The Annals of Statistics* (2025). Journal, arXiv.

7. Xuandong Zhao, Will Cai, Tianneng Shi, David Huang, **Licong Lin**, Song Mei, and Dawn Song. "Improving LLM safety alignment with dual-objective optimization." *International Conference on Machine Learning (ICML)* (2025). arXiv.

6. **Licong Lin**, Jingfeng Wu, Sham M. Kakade, Peter L. Bartlett, and Jason D. Lee. "Scaling laws in linear regression: Compute, parameters, and data." *Advances in Neural Information Processing Systems (NeurIPS)* (2024). arXiv.

5. Ruiqi Zhang*, **Licong Lin***, Yu Bai, Song Mei. "Negative preference optimization: From catastrophic collapse to effective unlearning." *Conference on Language Modeling (COLM)* (2024). arXiv.

4. **Licong Lin**, Yu Bai, and Song Mei. "Transformers as decision makers: Provable in-context reinforcement learning via supervised pretraining." *International Conference on Learning Representations (ICLR)* (2024). arXiv.

3. **Licong Lin** and Tijana Zrnic. "Plug-in performative optimization." *International Conference on Machine Learning (ICML)* (2024). arXiv.

2. **Licong Lin**, Mufang Ying, Suvrojit Ghosh, Koulik Khamaru, and Cun-Hui Zhang. "Statistical limits of adaptive linear models: Low-dimensional estimation and inference." *Advances in Neural Information Processing Systems (NeurIPS)* (2023). arXiv.

1. **Licong Lin** and Edgar Dobriban. "What causes the test error? Going beyond bias-variance via ANOVA." *Journal of Machine Learning Research (JMLR)* (2021). Journal, arXiv.

INVITED TALKS

- Statistics Seminar, Rutgers University, New Brunswick, NJ 02/2026
- Statistics & Data Science Seminar, Carnegie Mellon University, Pittsburgh, PA 01/2026
- Statistics Seminar, Cornell University, Ithaca, NY 01/2026
- Statistics & Data Science Seminar, UT Austin, Austin, TX 12/2025
- IDEAL workshop, Northwestern University, Evanston, IL 10/2025
- Symposium on Mathematical Foundations of Trustworthy Learning, Switzerland 10/2025
- Joint Statistical Meeting (JSM), Nashville, TN 08/2025
- INFORMS Annual Meeting, Seattle, WA 10/2024
- Joint Statistical Meeting (JSM), Portland, OR 08/2024
- UIUC Machine Learning Reading Group (online), Urbana, IL 04/2024